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Supplementary Materials for

Resonantly driven CNOT gate for electron spins

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Materials and Methods

The double quantum dot (DQD) device is fabricated from three layers of overlapping aluminum gates on top of a Si/SiGe quantum well heterostructure in natural Si (Figs. 1A and B). A Co micromagnet is deposited on top of the device to provide a slanting Zeeman field for spin manipulation (*30*). Two Agilent 33250 signal generators are used to control V_L and V_R throughout our experiments. Two Agilent 8267D microwave (MW) sources are used to drive gate S for performing single spin manipulation. To prevent unintentional microwave leakage from resonantly driving our system, each source was detuned from the electron spin resonance (ESR) frequency by 30 MHz. Using the Agilent 8267D internal IQ mixers, the baseband microwave frequencies were mixed up to the qubit frequencies using a Tektronix AWG5014C arbitrary waveform generator. A second Tektronix AWG5014C is used for rapid control of the exchange coupling via square pulses applied to gate M.

Supplementary Text

1. Micromagnet Design

The micromagnet design is shown in Fig. S1 and is based on recent work by Yoneda *et al.* (*30*). An external field B_{ext} = 0.5 T is applied along the z-axis to magnetize the cobalt film. There are in principle nine different components of the field gradient created by the micromagnet: dB_i/dj where $i, j = x, y, z$. The micromagnet is designed to maximize dB_z/dx and dB_y/dz. The dB_z/dx gradient gives rise to the different Zeeman splittings of the two qubits since they are spatially separated along the x-direction. The dB_y/dz gradient is intended to be the main mechanism for driving spin rotations. Using the voltage on gate S we oscillate the position of the electrons primarily in the z-direction, leading to an oscillating B_y magnetic field.

2. DQD Stability Diagram

The ground state charge configuration of the DQD is mapped out by monitoring the conductance of the charge sensor g_s as a function of the left and right plunger gate voltages V_L and V_R. The resulting charge stability diagram is shown in Fig. S2 where we plot the derivative of g_s with respect to V_L. The DQD is highly tunable and the (0,0) charge state is reached in the lower left corner of the charge stability diagram.

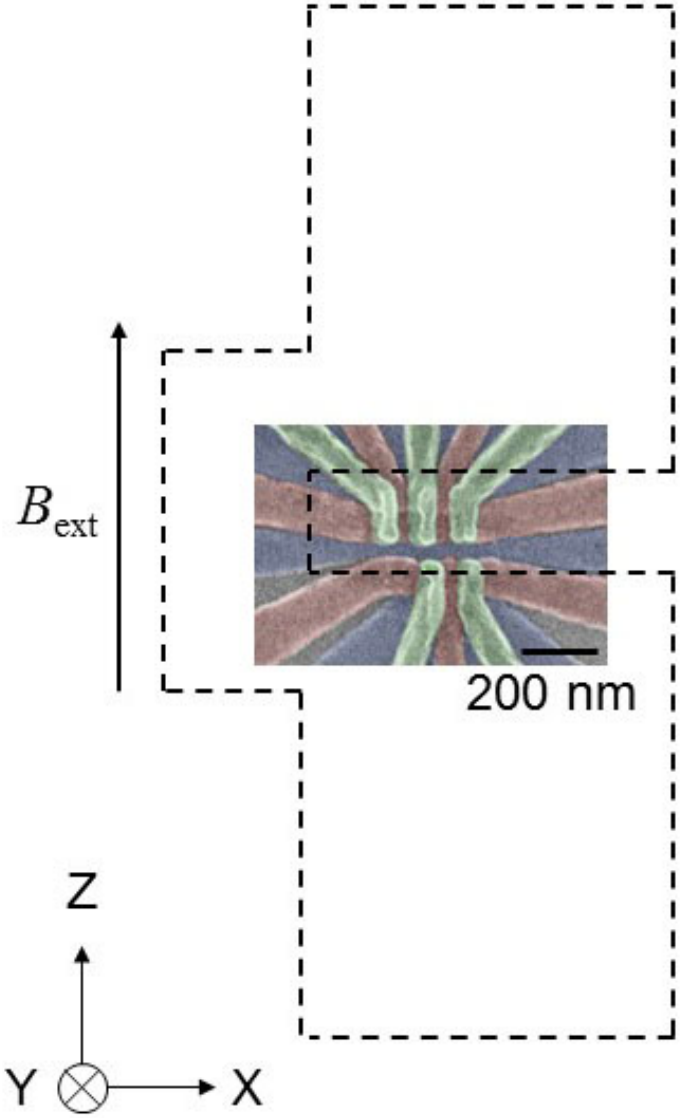

Fig. S1

False color scanning electron microscope image of the device with dashed lines denoting the region where Co is deposited. In these experiments an external field B_{ext} = 0.5 T is applied in the z-direction.

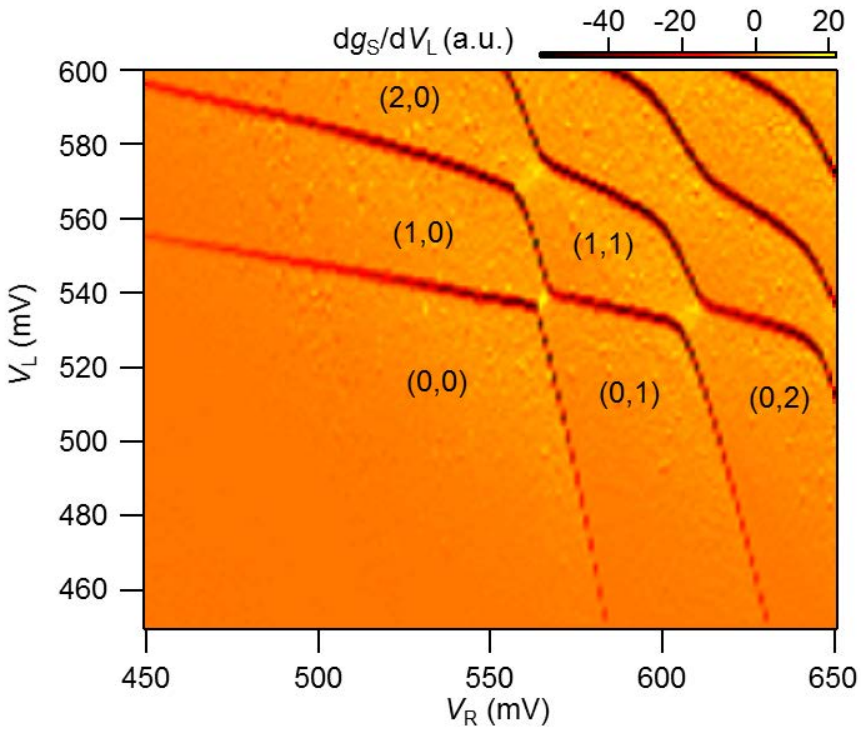

Fig. S2

DQD stability diagram. The derivative of g_s with respect to V_L is plotted as a function of V_L and V_R. The stable charge configurations are labelled by (N_L, N_R), where N_L (N_R) is the number of electrons in the left (right) dot.

3. Hamiltonian of a DQD in a Slanting Field

We operate our device in a regime where the difference of the Zeeman splitting in the two dots is much larger than the strength of the exchange interaction $\delta E_Z \gg J$ during all parts of the experiment. In this regime the exchange interaction will not drive a conventional SWAP operation due to the Zeeman energy difference of the left and right spins (*10*). Instead, the resonance frequency of each qubit becomes dependent on the state of the other qubit. We make use of this conditional single qubit frequency to directly drive a CNOT operation.

The DQD can be described in the basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ by the time dependent Hamiltonian

$$H(t) = J(t)\left(\boldsymbol{S}_\mathrm{L}\cdot\boldsymbol{S}_\mathrm{R} - \tfrac{1}{4}\right) + \boldsymbol{S}_\mathrm{L}\cdot\boldsymbol{B}_\mathrm{L} + \boldsymbol{S}_\mathrm{R}\cdot\boldsymbol{B}_\mathrm{R}. \tag{S1}$$

Here, the first term in the Hamiltonian describes the exchange interaction with strength $J(t)$ between the spin of the electron in the left dot $\boldsymbol{S}_\mathrm{L}$ and the spin of the electron in the right dot $\boldsymbol{S}_\mathrm{R}$. The second (third) term describes the influence of the magnetic field (in energy units), $\boldsymbol{B}_\mathrm{L} = \left(0, B^\mathrm{m}_\mathrm{y,L}(t), B^\mathrm{ext}_\mathrm{z} + B^\mathrm{m}_\mathrm{z,L}(t)\right)^T$ and $\boldsymbol{B}_\mathrm{R} =$

$\left(0, B_{\mathrm{y,R}}^{\mathrm{m}}(t), B_{\mathrm{z}}^{\mathrm{ext}} + B_{\mathrm{z,R}}^{\mathrm{m}}(t)\right)^{T}$, on the spin in the left (right) dot, composed of a homogeneous external magnetic field in the z-direction, $B_{\mathrm{z}}^{\mathrm{ext}}$ and an inhomogeneous field due to the micromagnet that contributes to both the y-component of each dot $B_{\mathrm{y,L}}^{\mathrm{m}}\left(B_{\mathrm{y,R}}^{\mathrm{m}}\right)$ and the z-component $B_{\mathrm{z,L}}^{\mathrm{m}}\left(B_{\mathrm{z,R}}^{\mathrm{m}}\right)$. We assume here that the magnetic field gradients are smaller than the single dot charging energies: $|\boldsymbol{B}_{\mathrm{L}} - \boldsymbol{B}_{\mathrm{R}}| \ll E_{\mathrm{C,L}}, E_{\mathrm{C,R}}$, where $E_{\mathrm{C,L}}, E_{\mathrm{C,R}}$ are the charging energies of both dots. In our experiment $E_{\mathrm{C,L}}, E_{\mathrm{C,R}}$ ~ 6 meV and $|\boldsymbol{B}_{\mathrm{L}} - \boldsymbol{B}_{\mathrm{R}}|$ ~ 1 μeV. Meunier *et al*. present a general Hamiltonian valid outside of this assumption in ref. (*31*).

We perform single qubit operations in a regime where J is approximately zero. Single-qubit gates are implemented by applying a sinusoidal drive to the S gate (shown in Fig. 1A of the main text). The oscillating electric field causes the positions of both electrons to oscillate in the fringing field of the micromagnet, which produces time-dependent out of plane magnetic fields in the DQD Hamiltonian:

$$B_{\mathrm{y,L}}^{\mathrm{m}}(t) = B_{\mathrm{y,L}}^{0} + B_{\mathrm{y,L}}^{1}\cos(2\pi f t + \varphi), \tag{S2}$$

$$B_{\mathrm{y,R}}^{\mathrm{m}}(t) = B_{\mathrm{y,R}}^{0} + B_{\mathrm{y,R}}^{1}\cos(2\pi f t + \varphi), \tag{S3}$$

where $B_{\mathrm{y,L}}^{0}$ and $B_{\mathrm{y,R}}^{0}$ are the static components of the out of plane fields determined by the average positions of the electrons, and $B_{\mathrm{y,L}}^{1}$ and $B_{\mathrm{y,R}}^{1}$ are the magnitudes of the oscillating fields determined by the amplitude of the microwave drive applied to gate S. By matching the microwave drive frequency f with the single spin resonance frequencies we can individually address each spin. The single spin rotation axis can be controlled by adjusting the phase φ of the microwave signal, producing an x rotation for $\varphi = -\pi/2$ and a y rotation for $\varphi = 0$.

4. Single Qubit Performance

Figures 2A and B of the main text show Rabi oscillations for each qubit, as extracted by measuring the spin-up probabilities as a function of the MW drive frequency and pulse length (with J = 0). The ESR frequencies differ by approximately 200 MHz due to the micromagnet field gradient. Driving the right qubit on resonance, we obtain Rabi oscillations that persist for at least 10 μs (Fig. 2C). The average probability differs from 0.5 due to the asymmetric readout fidelities of the spin-up and spin-down states.

We measure the qubit coherence times using Ramsey and Hahn echo pulse sequences. From the Ramsey decay we extract T_2^* = 1.2 μs for the left qubit and T_2^* = 1.4 μs for the right qubit. From the Hahn echo measurements we find T_2^{echo} = 22 μs for the left qubit and T_2^{echo} = 80 μs for the right qubit. The large difference in T_2^{echo} may be due to different field gradients seen by each qubit. A larger field gradient will lead to a stronger coupling of the qubit to charge noise and lead to faster dephasing. Further experiments are needed to elucidate the discrepancy.

In order to assess the single qubit gate fidelity we perform Clifford randomized benchmarking on both qubits (*32-34*). In Clifford randomized benchmarking *N* Clifford rotations are applied to rotate the spin qubit. At the end of the sequence a rotation is applied that would bring the qubit back to the spin-up state in the absence of any errors. The final spin state is then measured. The spin-up return probability $P_\uparrow = Ap_\mathrm{c}^\mathrm{N} + B$ is plotted as a function of the number of Cliffords *N* for both qubits in Figs. 2D and E. There are on average 1.875 single-qubit gates per Clifford gate (33). The spin-up probability is then fit by $P_\uparrow = Ap_\mathrm{c}^\mathrm{N} + B$ where p_c is the sequence fidelity. The Clifford fidelity can then be expressed in terms of the sequence fidelity as $F_\mathrm{c} = (1 + p_\mathrm{c})/2$. From these fits we obtain a Clifford fidelity of F_L = 99.3 ± 0.2 % and F_R = 99.7 ± 0.1 % for the left and right qubits, respectively.

5. Readout Fidelity from Numerical Simulations

To understand the limitations in our measurement fidelity and readout visibility, we have constructed a model similar to that used by Morello *et al.* (24). Our model can be summarized as follows:

1. Generate a series of time-domain current traces based on the measured tunneling rates Γ^on and Γ^off of both quantum dots.
2. Incorporate relaxation and loading errors by modifying an appropriate number of current traces.
3. Apply noise to the current traces using the measured noise spectrum.
4. Filter the current traces using the same software filters applied during the experiment and histogram the maximum currents.

5.1 Determining Tunneling Rates

To determine the tunneling rates, both qubits were prepared in the spin-up state and sequential single shot spin readout was performed 5000 times using the technique described by Elzerman *et al.* (*23*). The delay between the start of the readout and the rising edge of the current pulse is governed by the rate at which electrons tunnel off the quantum dot (Γ^off). By constructing a histogram of the rising edge times, one can fit the data to an exponential with a characteristic time constant Γ^off. Similarly, the falling edge of a given current pulse represents a tunneling-in event. By constructing a histogram of the width of the current pulses, we can extract the tunneling-in rate Γ^on. These data are plotted in Fig. S3.

5.2 Estimating the Readout Fidelity

Using the measured tunneling rates we generate simulated current traces for spin up and spin down electrons without noise. Based on our measured spin relaxation time T_1 = 22 ms and the simulated time before the electron hops out τ^off we convert a fraction of the simulated spin up traces to spin down with the following probability to simulate relaxation:

$$P_\mathrm{r} = e^{-\tau^\mathrm{off}/T_1}.$$

Similarly we convert a fraction of the spin down traces to spin up based on the calculated probability of unintentionally initializing into spin up based on our measured electron temperature $T_e \approx 150$ mK.

Next we add noise to the simulated traces using the measured noise spectrum of the charge sensor. Finally we filter the simulated current traces using the same filter applied in our experiment and histogram the peak-to-peak current ΔI_S measured during the readout window. The spin up readout fidelity $F_\uparrow$ and spin down readout fidelity $F_\downarrow$ are plotted as a function of the threshold current I_{th} in Fig. S4. The visibility is also calculated according to $V = 1 - F_\uparrow - F_\downarrow$ and plotted on the same graph. We find that given our experimental setup, our best expected visibility is 85% for the left dot and 78% for the right dot. This analysis is consistent with the maximum visibilities we have achieved in the experiment.

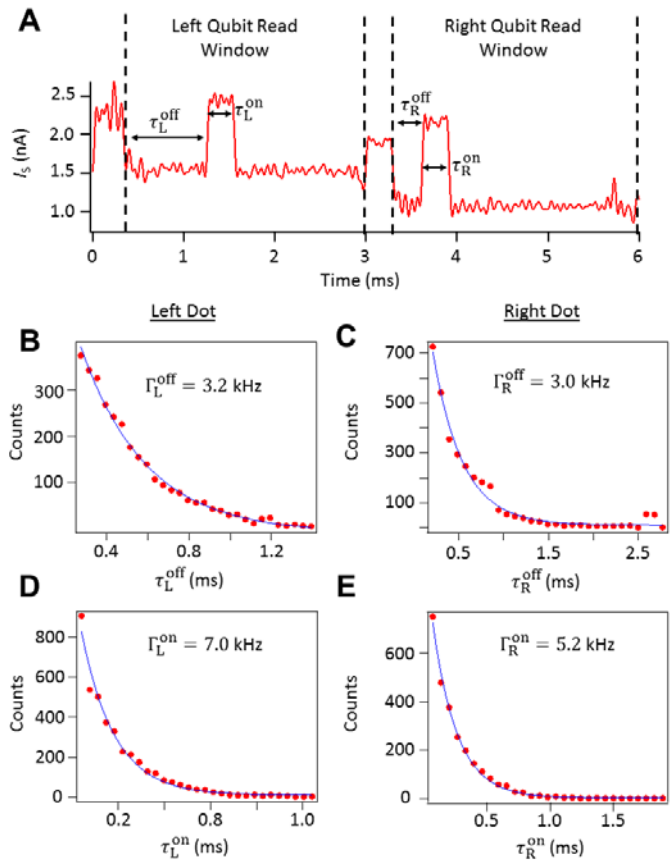

Fig. S3

Measuring tunneling rates during the readout process. An example single shot trace is shown in (**A**) where both electrons were detected in the spin up state. The square current blips at 1.3 and 3.7 ms are the spin signals. From these traces we can extract the time at which a spin up electron tunnels out and the time at which a spin down electron replaces it. We do this for 5,000 single shot traces and histogram the tunneling times to extract the four tunneling rates in (**B**)-(**E**).

5.3 Limitations to the Readout Fidelity

In these experiments, the electron temperature accounts for 10% of the reduction in our visibility, and spin relaxation accounts for the remaining 5-10%. In our experiment we readout the qubits sequentially, reading the left qubit first. Relaxation of the right qubit during the readout of the left qubit is what leads to the asymmetry of the expected visibilities. Relaxation contributions to the readout fidelity can be mitigated by using a faster readout technique such as RF reflectometry or by incorporating a cold amplifier into the readout circuit (*36*). Further improvements can be made by reducing the electron temperature or using cavity-based measurement approaches (*37*).

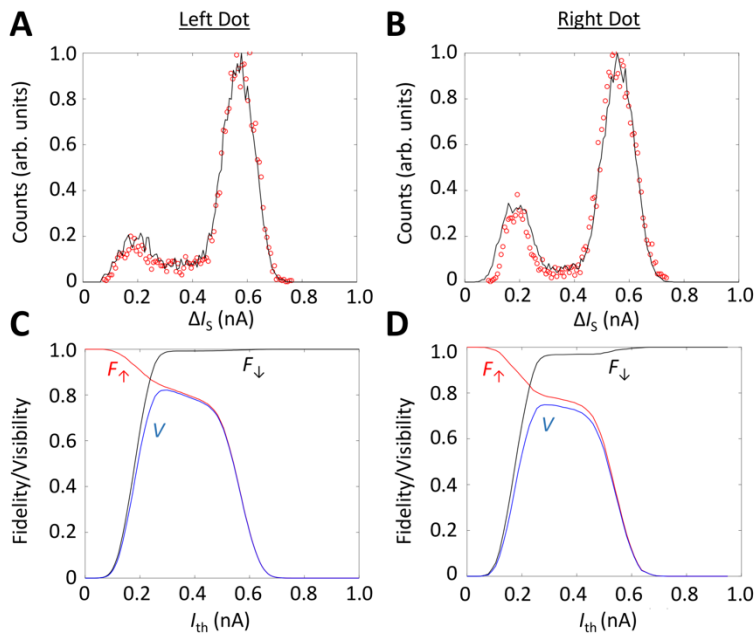

Fig. S4
Readout fidelities and visibilities for both quantum dots. Histograms of the peak-to-peak current ΔI_S during the read window for the left (**A**) and right quantum dots (**B**). Red dots are experimental data and black lines are simulation results. From the simulation results the fidelities for measuring the two spin states and readout visibility are plotted in (**C**) and (**D**) as a function of the current threshold I_{th}.

6. Exchange Interaction

The strength of the exchange interaction is experimentally adjusted by precisely controlling the voltage on the middle barrier gate V_M. Lowering (raising) the energy barrier between the dots increases (decreases) the transition matrix elements between the wavefunctions in the two dots. Additionally, changing the energy barrier between the dots also changes the position of the electrons in the field gradient thus adding a time-dependent contribution to the z-component of the magnetic fields $B_{z,Q}^{m}(t) = B_{z,Q}^{m,0} +$

$B_{\mathrm{z,Q}}^{\mathrm{m},1}(t)$ with $Q = L, R$. The two parameters $B_{\mathrm{z,L}}^{\mathrm{m},1}$ and $B_{\mathrm{z,R}}^{\mathrm{m},1}$ are responsible for the single-qubit frequency shifts that occur when exchange is turned on.
Setting $B_{\mathrm{y,L}}(t) = B_{\mathrm{y,R}}(t) = 0$ and defining the average Zeeman splitting $E_{\mathrm{Z}} = \left(B_{\mathrm{z}}^{\mathrm{ext}} + B_{\mathrm{z,L}}^{\mathrm{m},0} + B_{\mathrm{z}}^{\mathrm{ext}} + B_{\mathrm{z,R}}^{\mathrm{m},0}\right)/2$ and the splitting between the two qubits $\delta E_{\mathrm{Z}} = \left(B_{\mathrm{z}}^{\mathrm{ext}} + B_{\mathrm{z,R}}^{\mathrm{m},0} - B_{\mathrm{z}}^{\mathrm{ext}} - B_{\mathrm{z,L}}^{\mathrm{m},0}\right)$ we find for the instantaneous eigenvalues

$$\mathcal{E}(|\uparrow\uparrow\rangle) = E_{\mathrm{Z}} + \left(B_{\mathrm{z,L}}^{\mathrm{m},1} + B_{\mathrm{z,R}}^{\mathrm{m},1}\right)/2, \tag{S4}$$

$$\mathcal{E}\left(|\widetilde{\uparrow\downarrow}\rangle\right) = \left(-J - \sqrt{J^2 + \left(\delta E_{\mathrm{Z}} - \left(B_{\mathrm{z,L}}^{\mathrm{m},1} - B_{\mathrm{z,R}}^{\mathrm{m},1}\right)\right)^2}\right)/2, \tag{S5}$$

$$\mathcal{E}\left(|\widetilde{\downarrow\uparrow}\rangle\right) = \left(-J + \sqrt{J^2 + \left(\delta E_{\mathrm{Z}} - \left(B_{\mathrm{z,L}}^{\mathrm{m},1} - B_{\mathrm{z,R}}^{\mathrm{m},1}\right)\right)^2}\right)/2, \tag{S6}$$

$$\mathcal{E}(|\downarrow\downarrow\rangle) = -E_{\mathrm{Z}} - \left(B_{\mathrm{z,L}}^{\mathrm{m},1} + B_{\mathrm{z,R}}^{\mathrm{m},1}\right)/2. \tag{S7}$$

The tilde indicates the hybridization of the $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$ states due to the exchange interaction. These eigenvalues are plotted as a function of J in Fig. S5. As J is increased the antiparallel spin states are lowered in energy with respect to the parallel spin states. We find the corresponding transition frequencies

$$\begin{aligned} f_{|\psi_{\mathrm{R}}\rangle=|\downarrow\rangle}^{\mathrm{L}} &= \left|\mathcal{E}(|\downarrow\downarrow\rangle) - \mathcal{E}\left(|\widetilde{\uparrow\downarrow}\rangle\right)\right| \\ &= E_{\mathrm{Z}} + \left(-J + B_{\mathrm{z,L}}^{\mathrm{m},1} + B_{\mathrm{z,R}}^{\mathrm{m},1} - \sqrt{J^2 + \left(\delta E_{\mathrm{Z}} - \left(B_{\mathrm{z,L}}^{\mathrm{m},1} - B_{\mathrm{z,R}}^{\mathrm{m},1}\right)\right)^2}\right)/2, \end{aligned} \tag{S8}$$

$$\begin{aligned} f_{|\psi_{\mathrm{R}}\rangle=|\uparrow\rangle}^{\mathrm{L}} &= \left|\mathcal{E}\left(|\widetilde{\downarrow\uparrow}\rangle\right) - \mathcal{E}(|\uparrow\uparrow\rangle)\right| \\ &= E_{\mathrm{Z}} + \left(J + B_{\mathrm{z,L}}^{\mathrm{m},1} + B_{\mathrm{z,R}}^{\mathrm{m},1} - \sqrt{J^2 + \left(\delta E_{\mathrm{Z}} - \left(B_{\mathrm{z,L}}^{\mathrm{m},1} - B_{\mathrm{z,R}}^{\mathrm{m},1}\right)\right)^2}\right)/2, \end{aligned} \tag{S9}$$

$$\begin{aligned} f_{|\psi_{\mathrm{L}}\rangle=|\downarrow\rangle}^{\mathrm{R}} &= \left|\mathcal{E}(|\downarrow\downarrow\rangle) - \mathcal{E}\left(|\widetilde{\downarrow\uparrow}\rangle\right)\right| \\ &= E_{\mathrm{Z}} + \left(-J + B_{\mathrm{z,L}}^{\mathrm{m},1} + B_{\mathrm{z,R}}^{\mathrm{m},1} + \sqrt{J^2 + \left(\delta E_{\mathrm{Z}} - \left(B_{\mathrm{z,L}}^{\mathrm{m},1} - B_{\mathrm{z,R}}^{\mathrm{m},1}\right)\right)^2}\right)/2, \end{aligned} \tag{S10}$$

$$\begin{aligned} f_{|\psi_{\mathrm{L}}\rangle=|\uparrow\rangle}^{\mathrm{R}} &= \left|\mathcal{E}\left(|\widetilde{\uparrow\downarrow}\rangle\right) - \mathcal{E}(|\uparrow\uparrow\rangle)\right| \\ &= E_{\mathrm{Z}} + \left(J + B_{\mathrm{z,L}}^{\mathrm{m},1} + B_{\mathrm{z,R}}^{\mathrm{m},1} + \sqrt{J^2 + \left(\delta E_{\mathrm{Z}} - \left(B_{\mathrm{z,L}}^{\mathrm{m},1} - B_{\mathrm{z,R}}^{\mathrm{m},1}\right)\right)^2}\right)/2. \end{aligned} \tag{S11}$$

An important observation is that the shift in either electron's resonance frequency due to the state of the other electron exactly matches the exchange splitting

$$f^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\uparrow\rangle} - f^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\downarrow\rangle} = f^{\mathrm{R}}_{|\psi_{\mathrm{L}}\rangle=|\uparrow\rangle} - f^{\mathrm{R}}_{|\psi_{\mathrm{L}}\rangle=|\downarrow\rangle} = J. \tag{S12}$$

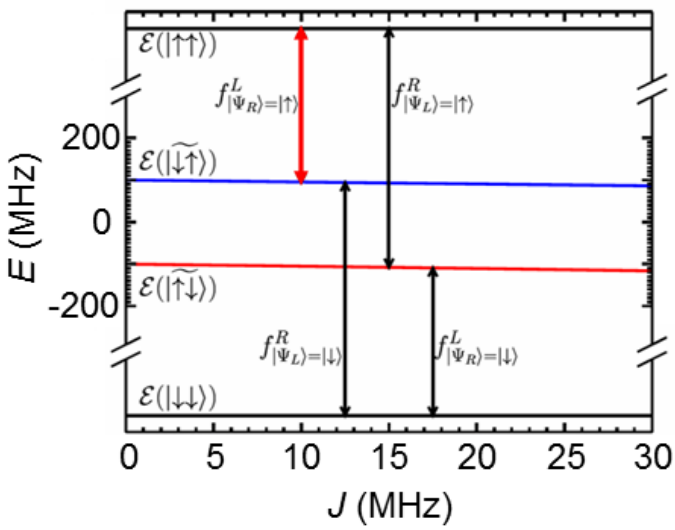

Fig. S5
Eigenenergies as a function of the exchange interaction J for the parameters in our experiment. The eigenstates are effectively $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ since the interaction between the two qubits is dominated by the large field gradient $\delta B \approx 200$ MHz $\gg J \approx$ 20 MHz. The black arrows indicate the relevant transitions in our experiment.

7. Exchange Spectroscopy

The pulse sequence used for the exchange spectroscopy of Fig. 3 is shown in Fig. S6. Both qubits are first initialized in spin-down. Then a microwave (MW) drive of length τ_{R} is applied to the right qubit preparing it in a state $|\psi\rangle_{\mathrm{R}} = (\cos(\theta/2)\,|\downarrow\rangle_{\mathrm{R}} - i\sin(\theta/2)\,|\uparrow\rangle_{\mathrm{R}})$ where θ is the rotation angle. Next, a low power drive of fixed length $\tau_{\mathrm{L}} \gg T_2$ is applied to the left qubit at frequency f_{p}. This low power probe tone is used to determine the resonance frequency of the left qubit, which will depend on the state of the right qubit due to exchange. When f_{p} is on resonance with the left qubit it will leave the

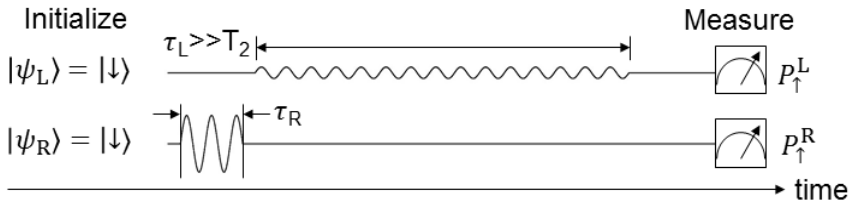

Fig. S6

Exchange spectroscopy pulse sequence. After initializing both qubits in spin-down a MW drive of length $\boldsymbol{\tau_R}$ is used to rotate the right qubit. Then a long ($\boldsymbol{\tau_L} \gg \boldsymbol{T_2}$) low power MW drive is applied to the left qubit. When on resonance, the left qubit is left in a mixed state. The resonance frequency of the left qubit in Fig. 3D is seen to depend on the rotation applied to the right qubit $\boldsymbol{\tau_R}$. This demonstrates that the resonance frequency of the left qubit depends on the state of the right qubit.

left qubit in a random state with 50% probability of being spin up. Finally both qubits are read out. The results of these measurements are shown in Figs. 3C and D of the main text. We observe that the left qubit resonance frequency alternates between two values, correlated with the oscillations in the spin state of the right qubit. This demonstrates that the left qubit resonance frequency is determined by the state of the right qubit, and also allows us to measure the exchange interaction as a function of the gate voltage V_M.

The exchange splitting can be approximated for t_c, $|\boldsymbol{B}_L - \boldsymbol{B}_R| \ll |E_C \pm \epsilon|$ by

$$J = \frac{2t_c^2(E_{C,L}+E_{C,R})}{(E_{C,L}+\epsilon)(E_{C,R}-\epsilon)}, \tag{S13}$$

with $E_{C,L}$ and $E_{C,R}$ being the charging energy of the left and right quantum dots, and t_c being the tunneling matrix element between the dots. The tunneling matrix element depends on the middle barrier potential V_M and can be approximated as $t_c = E\tau$ where τ denotes the tunneling amplitude and E the single-dot confinement energy. Then $J \propto t_c^2 \propto |\tau|^2$. As a simple model, we use the tunneling probability through a rectangular barrier in the limit of weak tunneling

$$|\tau|^2 = \frac{16E(V-E)}{V^2} exp\left(-2W\sqrt{2m|V-E|/\hbar^2}\right), \tag{S14}$$

where V and W denote the potential barrier height and width and m the electron mass. Using $V \propto -V_M + const.$ we find

$$J(V_M) = c\frac{V_{M0}-V_M}{(V_M-V_{M1})^2} exp\left(-\sqrt{|V_M - V_{M0}|/V_{on}}\right), \tag{S15}$$

Where V_{M1} is the voltage at which the tunneling barrier height is zero, V_{M0} is the voltage at which the barrier height equals the electron energy, V_{on} describes the voltage scale of the sub-exponential increase of J with V_M, and c is an overall scale factor. By fitting the experimentally determined values of J to this function we find $V_{on} = 0.559$ mV, $V_{M0} = 412.8$ mV, $V_{M1} = 451.8$ mV and $c = 16000$ (GHz)^2/e. The measured $J(V_M)$ is in good agreement with the predictions of this simple model (Fig. S7).

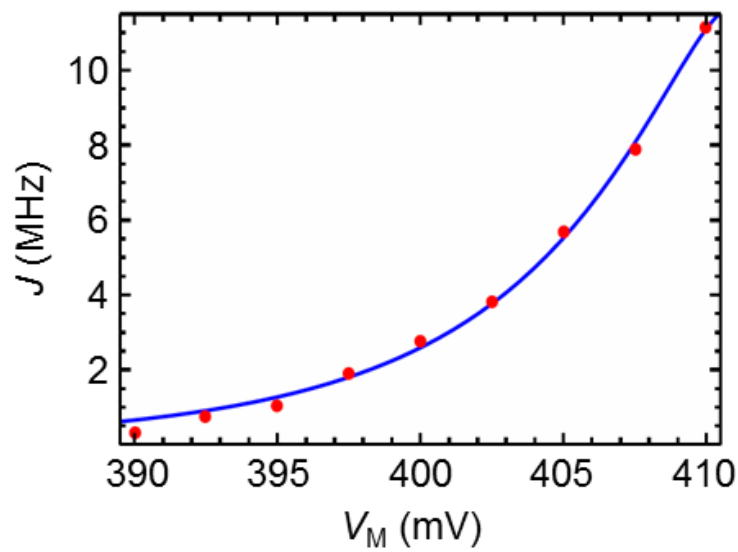

Fig. S7
J as a function of V_{M} (dots), measured using the pulse sequence of Fig. 3. The blue curve is a fit to the data based on tunneling through a rectangular barrier.

8. CNOT Gate

A single-shot two-qubit CNOT gate is implemented by applying a square pulse to the M-gate to turn on the exchange interaction for a time τ_{dc} while simultaneously driving the S-gate at the transition frequency $f^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\uparrow\rangle}$ for a time τ_{P}. The square pulse must be applied adiabatically with respect to the Zeeman splitting of the two qubits so 5 ns linear voltage ramps are used whenever the voltage is changed on the M-gate. The resonant driving results in coherent rotations between the $|\widetilde{\downarrow\uparrow}\rangle$ and $|\uparrow\uparrow\rangle$ states while the other states stay untouched. This results in a CNOT gate with the right qubit as the control and the left qubit as the target.

8.1 DC Exchange Dynamics

We experimentally measure the four transition frequencies when J is nonzero by making use of a pulsed electron spin resonance technique sensitive to brief frequency shifts (*38*). To measure the change in frequency of the left qubit when exchange is turned on ($\Delta f_{L,ex} = f^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\downarrow\rangle} - f^{\mathrm{L}}_{J=0}$) we start with the state $|\psi_{\mathrm{in}}\rangle = |\downarrow\downarrow\rangle$ and perform a Hahn echo pulse sequence on the left qubit. By applying an exchange pulse, we detune the left qubit by an amount $\Delta f_{L,ex}$ and if the exchange pulse is only applied in the first half of the Hahn echo sequence for a duration τ_{dc}, a phase is accumulated which is equal to $\tau_{dc}\Delta f_{L,ex}$. By measuring the spin up probability of the left dot as a function of the length

of the exchange pulse τ_{dc}, we can track the phase acquired by the left qubit and extract the frequency shift as shown in Fig S8. By repeating this experiment on the left qubit with the input states $|\psi_{in}\rangle = |\downarrow\downarrow\rangle, |\downarrow\uparrow\rangle$ and the right qubit with the input states $|\psi_{in}\rangle = |\downarrow\downarrow\rangle, |\uparrow\downarrow\rangle$ we can map out the frequency shift in the two qubits arising from exchange.

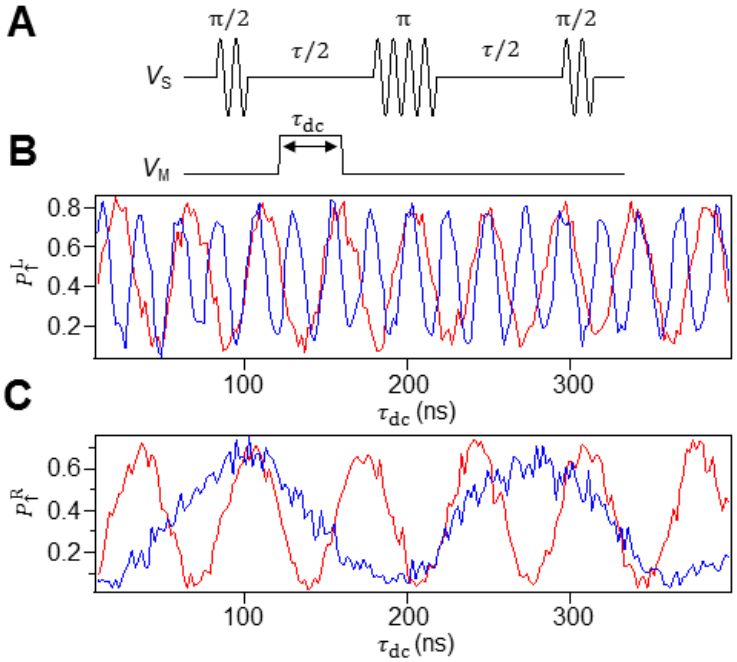

Fig. S8

Measuring single qubit transition frequencies with a Hahn echo sequence. (A) We apply a Hahn echo pulse sequence to one of the qubits with τ = 2 µs. By applying a dc exchange pulse during only one period of free evolution during the echo sequence we accumulate phase proportional to the frequency shift induced by the dc exchange pulse. (B) $\boldsymbol{P}_\uparrow^{\mathbf{L}}$ as a function of τ_{dc} for the right qubit in the spin up state (blue trace) and spin down state (red trace). (C) $\boldsymbol{P}_\uparrow^{\mathbf{R}}$ as a function of τ_{dc} for the left qubit in the spin up state (blue trace) and spin down state (red trace).

The exchange giving rise to the frequency shift is an entangling operation so to obtain a pure CNOT, it is critical that we appropriately cancel out this interaction. To see how we cancel this interaction, let us focus only on the phases of the left qubit; preparing both spins in a superposition we start with the initial wavefunction

$$\begin{aligned} |\psi(0)\rangle &= (|\downarrow\rangle_L + |\uparrow\rangle_L) \otimes \frac{(|\downarrow\rangle_R + |\uparrow\rangle_R)}{2} \\ &= (|\downarrow\rangle_L + |\uparrow\rangle_L) \otimes |\downarrow\rangle_R/2 + (|\downarrow\rangle_L + |\uparrow\rangle_L) \otimes |\uparrow\rangle_R/2. \end{aligned} \tag{S16}$$

Consider the evolution of the wavefunction after the exchange is adiabatically turned on (relative to the Zeeman term in the spin Hamiltonian) and held at a value J for a time τ_{dc}. Looking at only the phase acquisition on the left dot, the wavefunction after a time τ_{dc} evolves into

$$\begin{aligned}|\psi(t)\rangle = &\left(|\downarrow\rangle_{\mathrm{L}} + e^{-if^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\downarrow\rangle}\tau_{\mathrm{dc}}}|\uparrow\rangle_{\mathrm{L}}\right)\otimes\frac{|\downarrow\rangle_{\mathrm{R}}}{2}\\ &+\left(|\downarrow\rangle_{\mathrm{L}} + e^{-if^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\downarrow\rangle}\tau_{\mathrm{dc}}}e^{-iJ\tau_{\mathrm{dc}}}|\uparrow\rangle_{\mathrm{L}}\right)\otimes|\uparrow\rangle_{\mathrm{R}}/2,\end{aligned} \tag{S17}$$

due to the property $f^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\uparrow\rangle} = f^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\downarrow\rangle} + J$ from Eq. (S12). Thus for $\tau_{\mathrm{dc}} = 2\pi/J$ the two-qubit entangling operation is effectively canceled out so that the left dot will have the same phase regardless of the state of the right dot. The remaining single qubit phase is determined solely by $f^{\mathrm{L}}_{|\psi_{\mathrm{R}}\rangle=|\downarrow\rangle}$. The same argument holds for the right qubit.

8.2 AC Gate

Due to the choice of $\tau_{\mathrm{dc}} = 2\pi/J$ the conditional phases accumulated by the dc exchange pulse are completely cancelled out. An additional ac gate is applied to resonantly drive a spin transition between $|\widetilde{\downarrow\uparrow}\rangle$ and $|\uparrow\uparrow\rangle$. This results in a CNOT gate with the following truth table:

$$\begin{aligned}|\uparrow\uparrow\rangle &\rightarrow |\downarrow\uparrow\rangle,\\ |\uparrow\downarrow\rangle &\rightarrow |\uparrow\downarrow\rangle,\\ |\downarrow\uparrow\rangle &\rightarrow |\uparrow\uparrow\rangle,\\ |\downarrow\downarrow\rangle &\rightarrow |\downarrow\downarrow\rangle.\end{aligned} \tag{S18}$$

To support our experiment we numerically simulate the unitary time evolution of our two qubit system. We integrate the time-dependent Schrodinger equation $i\hbar\dot{\Psi}(t) = H(t)\Psi(t)$ in the rotating frame of the two microwave sources $\widetilde{H}(t) = U^{\dagger}HU + iU^{\dagger}U$ with $U = exp\left[i\omega t\left(S_{\mathrm{z,L}} + S_{\mathrm{z,R}}\right)/\hbar\right]$ to cancel out the fast oscillations from the time-independent magnetic field gradient terms. In Fig. S9 we numerically simulate the pulse sequence for the frequency selective microwave gate detailed in Fig. 4 of the main text. The spin-up probabilities for the two input states $|\psi_{\mathrm{in}}\rangle = |\uparrow\uparrow\rangle$ and $|\psi_{\mathrm{in}}\rangle = |\downarrow\uparrow\rangle$ show anti-correlated oscillations. For τ_{p} = 130 ns a CNOT is realized, in good agreement with the experimental data.

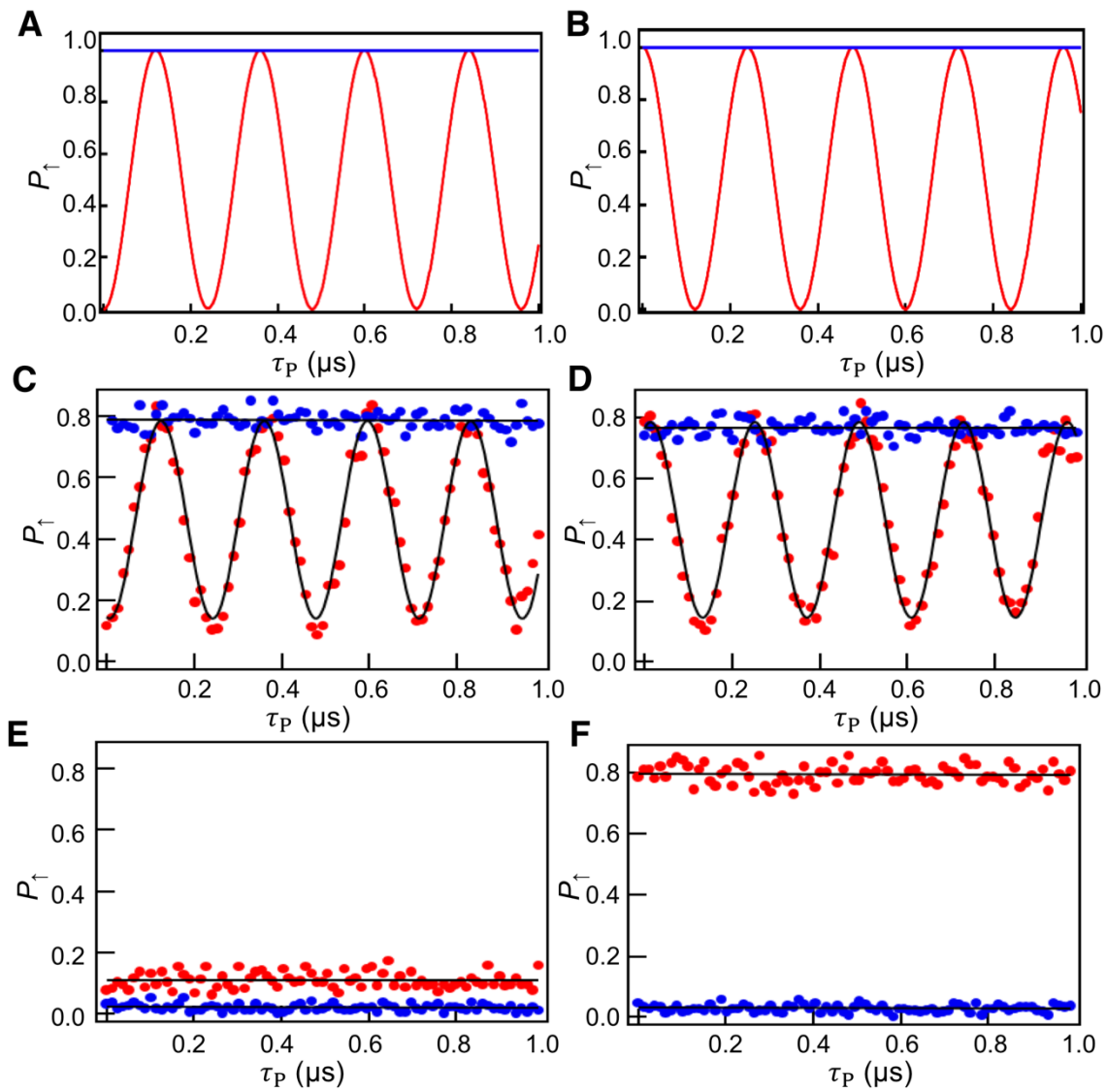

Fig. S9
Simulated conditional oscillations from the CNOT pulse sequence by time integration of Eq. 1 (S*28*). Simulated spin up probabilities for an input state $|\boldsymbol{\psi}_{\mathbf{in}}\rangle = |\downarrow\uparrow\rangle$ (**A**) and for $|\boldsymbol{\psi}_{\mathbf{in}}\rangle = |\uparrow\uparrow\rangle$ (**B**). Measured spin up probabilities for $|\boldsymbol{\psi}_{\mathbf{in}}\rangle = |\downarrow\uparrow\rangle$ (**C**), $|\boldsymbol{\psi}_{\mathbf{in}}\rangle = |\uparrow\uparrow\rangle$ (**D**), $|\boldsymbol{\psi}_{\mathbf{in}}\rangle = |\downarrow\downarrow\rangle$ (**E**), and $|\boldsymbol{\psi}_{\mathbf{in}}\rangle = |\uparrow\downarrow\rangle$ (**F**). The measured oscillation amplitudes differ from the simulations due to limited readout visibility. The black curves are least squares fits to the data.

9. State Tomography of a Bell State

We now use the CNOT gate to prepare and readout a Bell state. By applying a π/2 pulse on the right qubit (control) we prepare the input state $|\psi_{\text{in}}\rangle = (|\downarrow\downarrow\rangle - i|\downarrow\uparrow\rangle)/\sqrt{2}$. After applying the CNOT gate, the resulting wave function is the Bell state $|\psi_{\text{out}}\rangle = (|\downarrow\downarrow\rangle - i|\uparrow\uparrow\rangle)/\sqrt{2}$. We then use single qubit gates to perform full two-qubit state

tomography on the output state. The measured density matrix has a fidelity $F = \langle\psi_{\text{target}}|\rho|\psi_{\text{target}}\rangle = 56.7$ %. This fidelity has contributions from both the finite readout visibility and errors in the two qubit gate. We therefore correct for readout errors using the techniques described below and extract a two-qubit gate fidelity of 78.1 %.

The readout fidelities are determined from the data shown in Fig. 4E of the main text. Fits to the oscillations of the left and right spin probabilities yield fidelities of $f_{L,\downarrow}$= 0.888, $f_{L,\uparrow}$= 0.793, $f_{R,\downarrow}$= 0.969, $f_{R,\uparrow}$= 0.786. Using these readout fidelities we recalculate the density matrix. The raw is a set of 15 measurements $P^M = \left(P^M_{|\downarrow\downarrow\rangle}, P^M_{|\downarrow\uparrow\rangle}, P^M_{|\uparrow\downarrow\rangle}, P^M_{|\uparrow\uparrow\rangle}\right)$ on the set of two-qubit Pauli operators
$M = \{IX, IY, IZ, XI, Y\,I, ZI, XX, XY, XZ, YX, YY, YZ, ZX, ZY, ZZ\}$. We apply the transformation defined in (*22*) to our raw data to remove readout errors from our fidelity analysis. For each spin we relate the measured probabilities P^M to the actual two-spin probabilities $P^M = \left(P^M_{|\downarrow\downarrow\rangle}, P^M_{|\downarrow\uparrow\rangle}, P^M_{|\uparrow\downarrow\rangle}, P^M_{|\uparrow\uparrow\rangle}\right)$ using the equations from Watson *et al.* (22):

$$P^M = \left(\hat{F}_L \otimes \hat{F}_R\right)P, \tag{S19}$$

where,

$$\hat{F}_i = \begin{pmatrix} F_{i,\downarrow} & 1 - F_{i,\uparrow} \\ 1 - F_{i,\downarrow} & F_{i,\uparrow} \end{pmatrix}. \tag{S20}$$

Finally we construct the density matrix shown in Fig. S10 using the set of calculated two-spin probabilities P and a maximum-likelihood estimate as outlined in (*38-39*). The corrected density matrix has a Bell state fidelity of 78.1%.

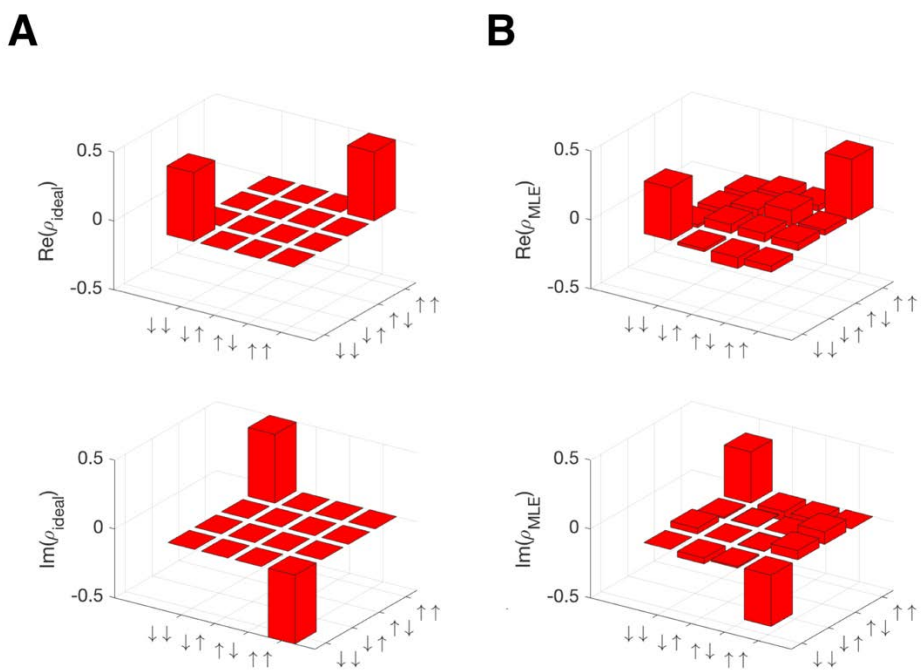

Fig. S10
(**A**) Expected density matrix of the ideal Bell state including no SPAM errors. (**B**) Experimentally determined density matrix constructed using a maximum-likelihood estimate. The fidelity F = 78%.

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